**Lab 2: IT Support & IAM Ticket Handling (Active Directory)**

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## Objective

Simulate real-world IT Service Desk and Identity & Access Management (IAM) scenarios using Active Directory. This lab demonstrates hands-on handling of common user account tickets, troubleshooting authentication issues, applying security policies, and documenting learning outcomes suitable for a professional IT portfolio.

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## Environment

- Hypervisor: VirtualBox
- Virtual Machine: Windows Server (Domain Controller)
- Domain: LAB (lab.local)
- Administrator Account: LAB\Administrator
- Users Involved:
  - joejane (standard domain user)
  - [second user]
- Security Group:
  - SOCadmin
- Tools Used:
  - Server Manager
  - Active Directory Users and Computers (ADUC)
  - Group Policy Management
  - Command Prompt

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## Ticket 1: User Cannot Log In (Password Issue)

### Issue Description

User reported inability to log in due to incorrect username or password.

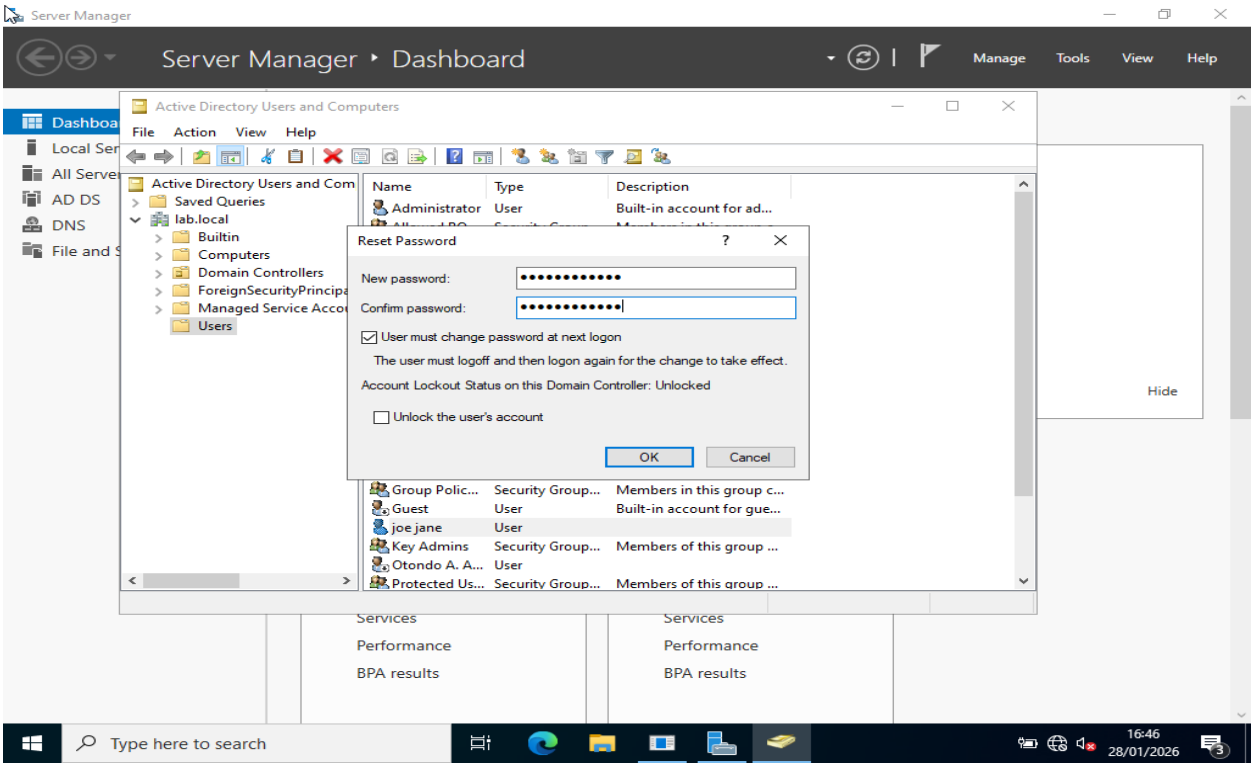
### Investigation

- Verified user account exists in ADUC
- Checked account status:
  - Account disabled: No
  - Account locked: No

## Resolution

- Reset password for user **joejane**
- Provided temporary password
- Confirmed password reset confirmation

**Screenshot:** Lab2_Ticket1_PasswordReset.png

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# Ticket 2: User Account Disabled / Access Issue

## Issue Description

User received message indicating account was disabled.

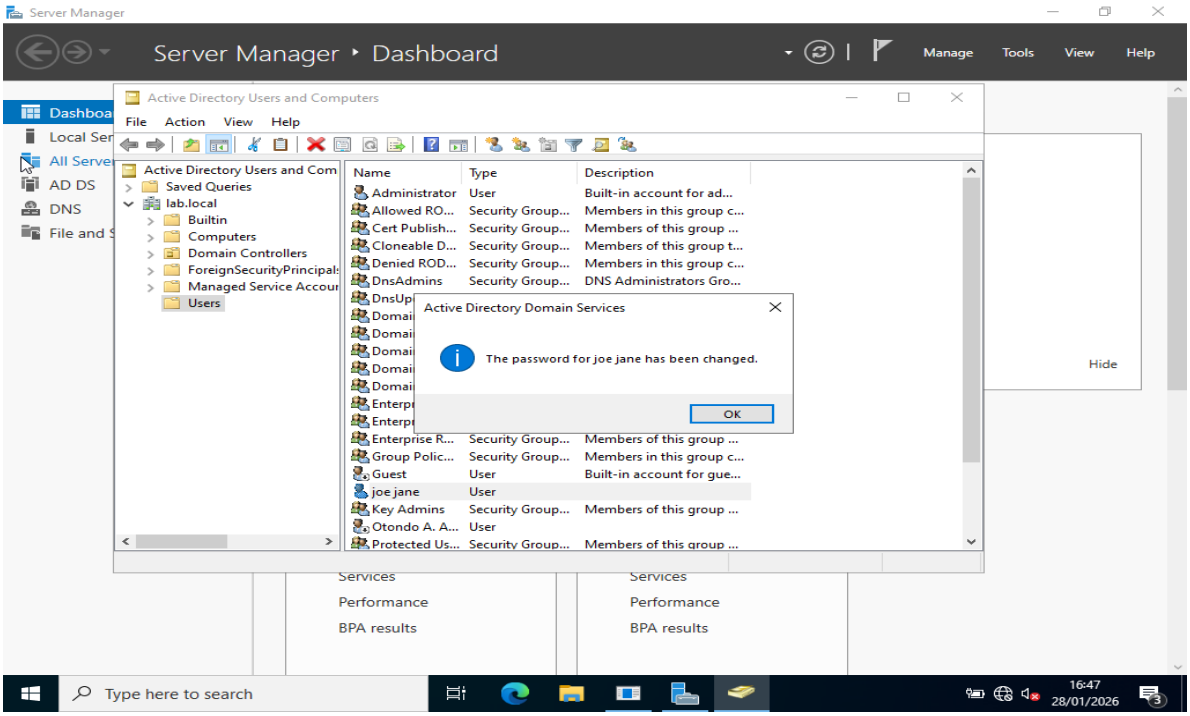
## Investigation

- Opened ADUC → User Properties
- Verified account status icon and settings

## Resolution

- Confirmed account was enabled
- No further action required

**Screenshot:** Lab2_Ticket2_AccountStatus.png

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# Ticket 3: Group Membership / Privileged Access Request

## Issue Description

User requested elevated access via security group membership.

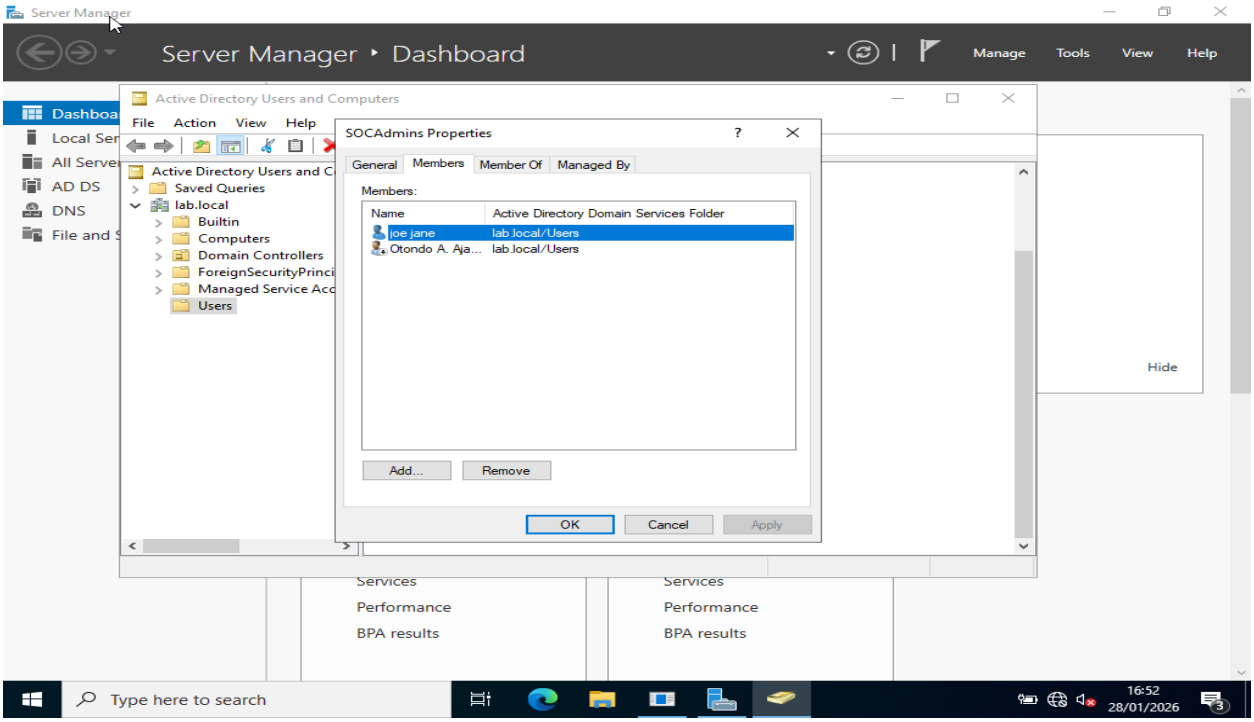
## Investigation

- Opened SOCadmin group properties
- Reviewed current members

## Resolution

- Added **joejane** and a second user to **SOCadmin** group
- Verified group membership

**Screenshot:** Lab2_Ticket3_SOCadminGroup.png

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# Ticket 4: Account Lockout (Advanced IAM Scenario)

## Issue Description

User unable to authenticate after multiple failed login attempts.

## Initial Observation

- Attempts to simulate lockout failed
- Account did not lock despite repeated authentication failures

## Root Cause

No account lockout policy configured in the domain (default setting).

## Action Taken

### *Configure Account Lockout Policy*

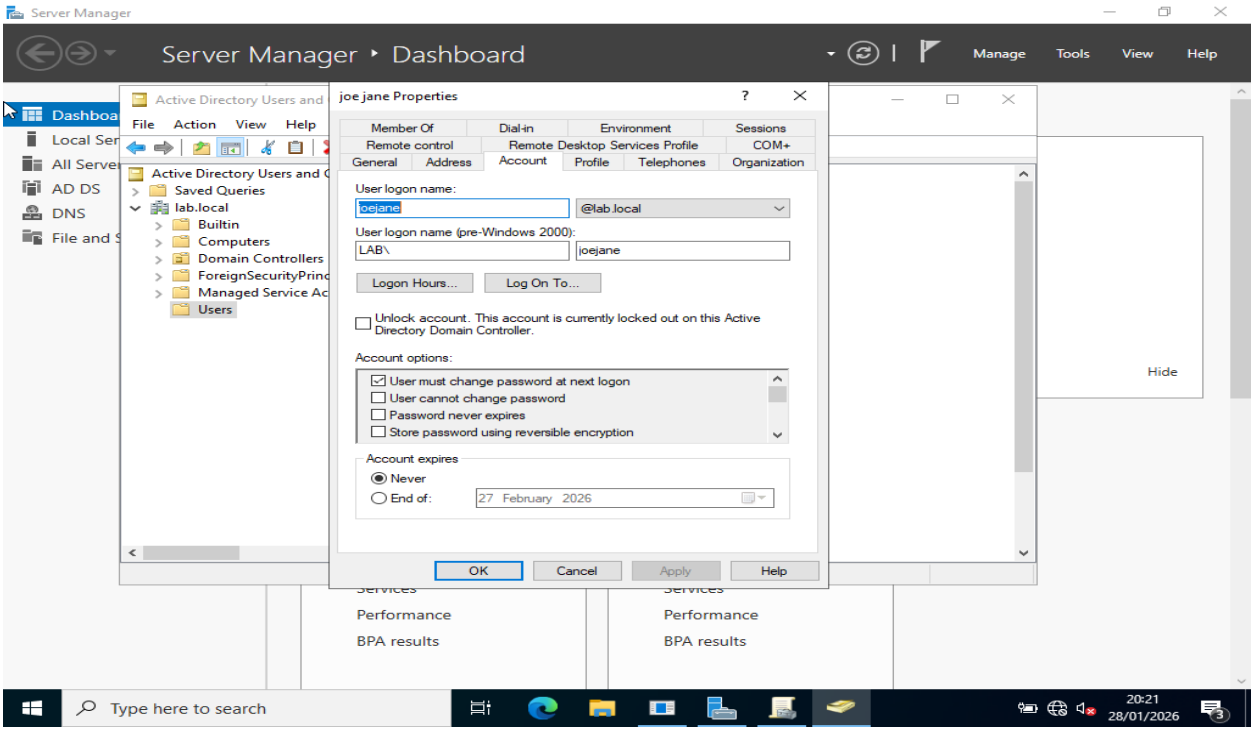
1. Opened Group Policy Management
2. Edited **Default Domain Policy**
3. Navigated to: Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Account Lockout Policy
4. Configured:
    - Account lockout threshold: 3
    - Account lockout duration: 30 minutes

- Reset account lockout counter after: 30 minutes

5. Forced policy update using:

```
gpupdate /force
```

**Screenshot:** Lab2_Ticket4_LockoutPolicy.png

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## Simulating Lockout

- Used command:

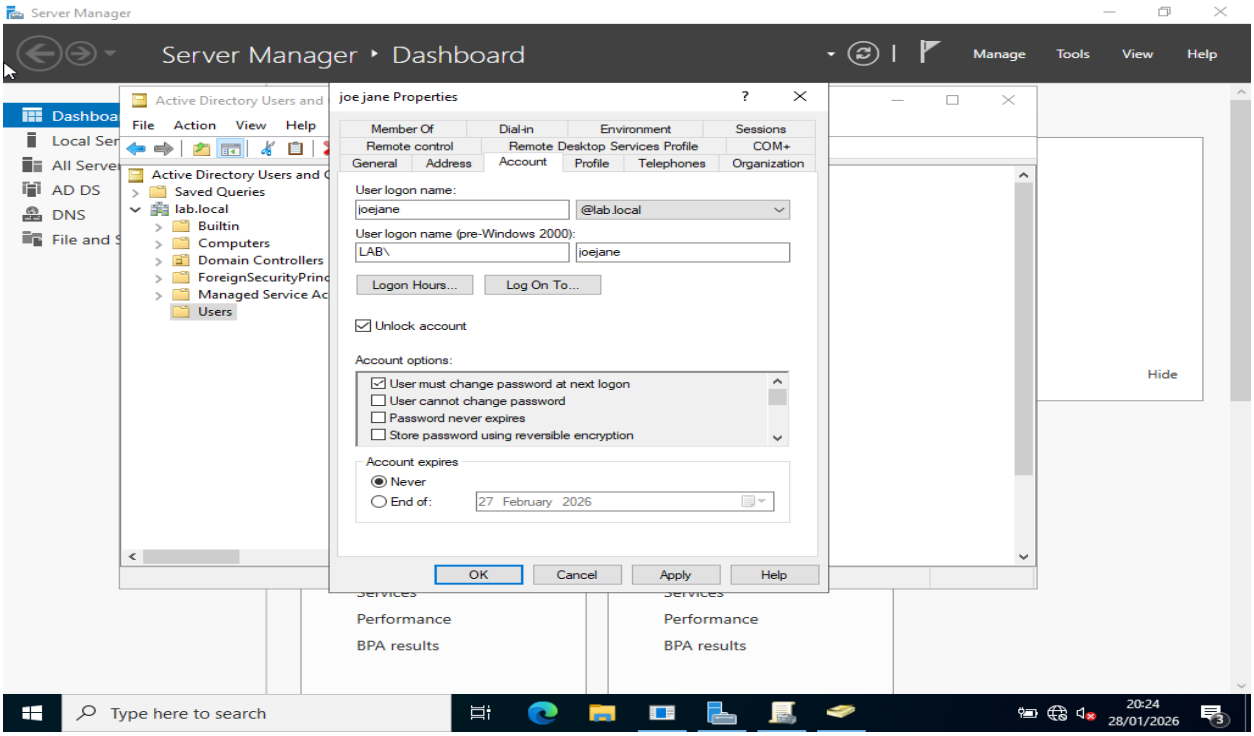
```
runas /user:LAB\\joejane cmd
```

- Entered incorrect password multiple times
- Received confirmation: “The referenced account is currently locked out and may not be logged on”

## Verification

- Checked user properties in ADUC
- Confirmed **Account is locked out** checkbox enabled

**Screenshot:** Lab2_Ticket4_AccountLocked.png

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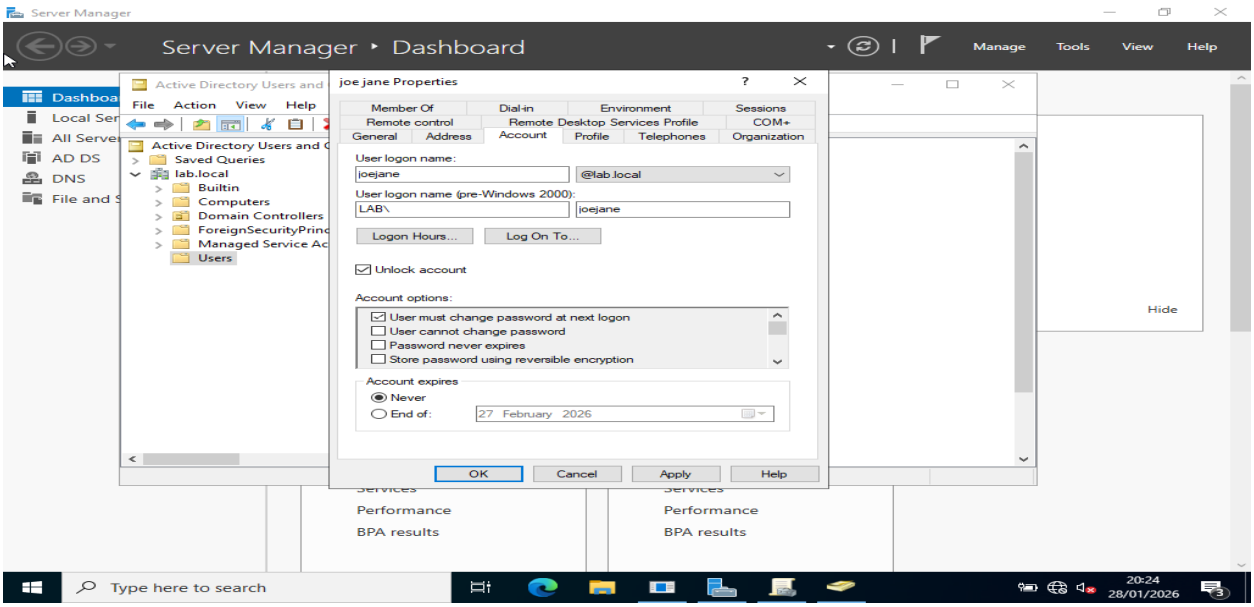
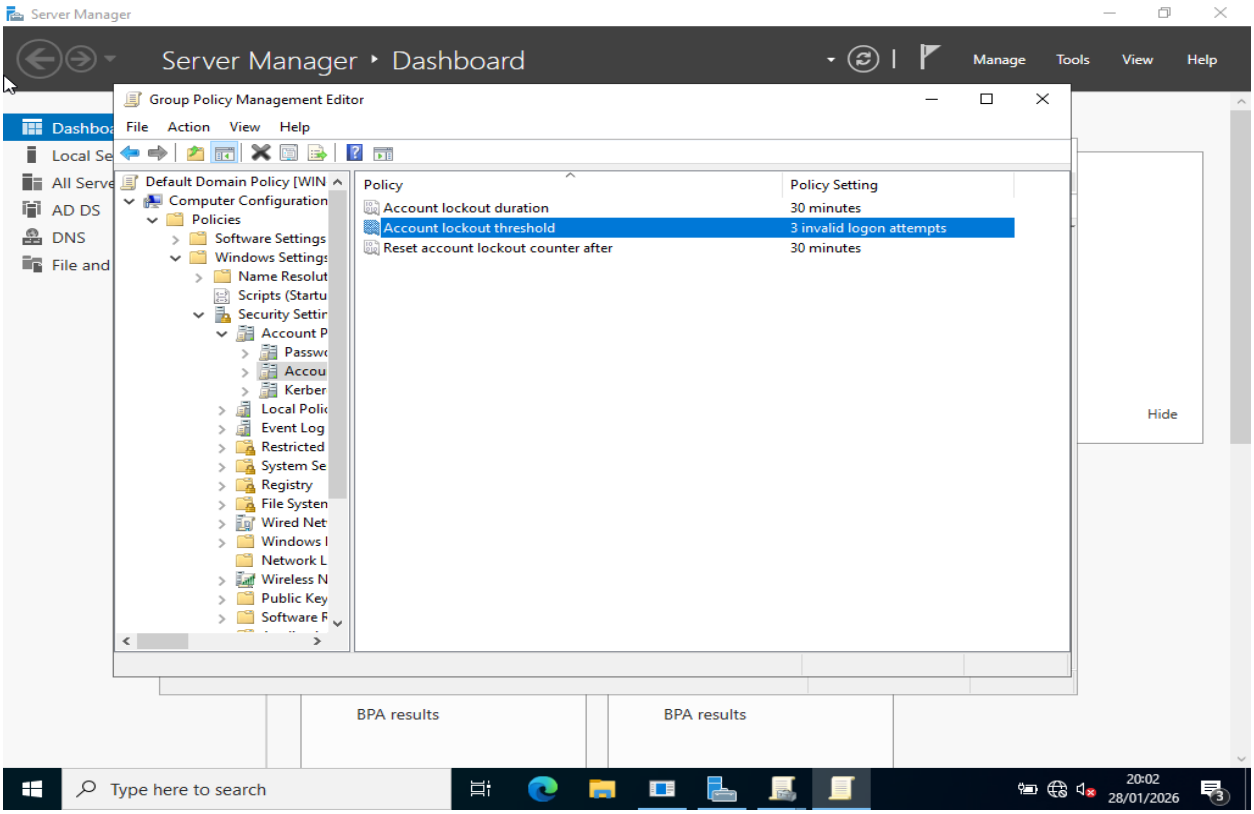
## Resolution

- Unchecked **Account is locked out** in ADUC
- Applied changes
- User account restored

**Screenshot:** Lab2_Ticket4_AccountUnlocked.png



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## Challenges & Troubleshooting Highlights

- Encountered System Error 53 when using `net use` due to name resolution issues
- Learned multiple methods exist to simulate authentication failures
- Understood default domain policies may not enforce security controls until configured
- Identified differences between Domain Controller login behavior and standard client machines
- Managed VM keyboard capture and focus issues during Ctrl+Alt+Del operations

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## Lessons Learned

- IAM troubleshooting requires policy awareness, not just account checks
- Group Policy plays a critical role in enterprise security
- Account lockouts are policy-driven, not automatic
- Clear documentation is essential for IT support roles
- Real-world IT involves diagnosing both technical and configuration-related issues

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## Skills Demonstrated

- Active Directory user and group management
- Password resets and account recovery
- Group Policy configuration
- Account lockout simulation and resolution
- Troubleshooting authentication and access issues
- Professional IT documentation

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## Next Steps

- Lab 3: Windows Server Services & Event Viewer
- Lab 4: Client VM Join to Domain
- Lab 5: Azure AD / Microsoft Entra ID Basics

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**Note on screenshots:** Insert screenshots manually where indicated. Suggested filenames: - Lab2_Ticket1_PasswordReset.png - Lab2_Ticket2_AccountStatus.png - Lab2_Ticket3_SOCadminGroup.png - Lab2_Ticket4_LockoutPolicy.png - Lab2_Ticket4_AccountLocked.png - Lab2_Ticket4_AccountUnlocked.png